

Weight stability intervals for multi-criteria decision analysis using the weighted sum model

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ABSTRACT

Multi-criteria decision analysis (MCDA), or multi-criteria decision making (MCDM) encompasses a wide set of mathematical methods to compare alternatives whilst considering multiple criteria and the relative importance (weight) of each criterion or the allowable trade-off between criteria. MCDA/MCDM may be implemented in a decision support system to aid decision makers compare multiple alternatives. A key consideration in any MCDA is the sensitivity of the results obtained to alterations of the criteria weights used. Prior methods to determine the weight stability interval (WSI, the range of values for individual weights for which the ranking of alternatives will not change) include the manual alteration of criteria weights, the use of different sets of criteria weights in a form of scenario analysis, or the process of iteratively searching for the maximum and minimum allowable alterations to individual criteria weights to determine the weight stability interval. The first contribution of this paper is the development of a novel method for determining precise weight stability intervals which does not rely on enumeration or simulation for use with the weighted sum model (WSM, based on the L_1 Minkowski norm), one of the most widely used aggregation methods within MCDA. The second contribution of this work is the use of these WSIs to identify the most sensitive weight(s) based on a Pareto dominance approach whilst considering both the allowable increase and decrease in criteria weights. The above points are demonstrated and applied in an analysis of several alternatives for renewable energy (biogas) production.

1. Introduction

1.1. Research background

Multi-criteria decision making or multi-criteria decision analysis (MCDM/MCDA) encompasses a wide variety of mathematical methods to aid in the selection of the best or optimal alternative from a list of available alternatives based on the performance of the alternatives with regards to relevant criteria in a systematic approach. Other goals of such methods are to rank all available alternatives, sort them into groups, and determine their relative priorities when all criteria are considered simultaneously.

The process of decision making involves choices among alternatives, be they projects to implement or actions to take, and we make decisions by comparing these alternatives to one another with a goal of choosing the one with the most desirable outcome (Kossiakoff et al., 2020). MCDM/MCDA approaches are at the core of decision support systems (DSS), an information system that supports decision-making activities. Thus, it is only natural that MCDM/MCDA and DSSs have received plenty of attention by the systems community as indicated by the following recent publications in the journal of Expert Systems with Applications (Barrera et al., 2024; Dweiri et al., 2016; Ma & Li, 2024; Solares et al., 2022).

There is a plethora of MCDA methods available in the literature

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which incorporate different normalisation, weighting, and aggregation steps. Common aggregation methods include compensatory trade-off methods such as simple additive weighting (SAW) also known as the weighted sum model (WSM) (Fishburn, 1967), compromise programming (Zeleny, 1974) (Yu, 1973), simple multi attribute rating technique (SMART) (Edwards, 1977), Technique of Order Preference by Similarity to the Ideal Solution (TOPSIS) (Hwang & Yoon, 1981), and VlseKriterijumska Optimizacija I Kompromisno Resenje (VIKOR) (Opricovic & Tzeng, 2002); outranking methods such as ELimination Et Choix Traduisant la REalite (ELECTRE) (Roy, 1968) and Preference Ranking Organization Method for Enrichment Evaluations (PROMETHEE) (Brans et al., 1986), pairwise comparison methods such as the analytical hierarchy process (AHP) (Saaty, 1977), and stochastic methods such as stochastic multi-objective acceptability analysis (Lahdelma et al., 1998).

Recent works have proposed a unified taxonomy of MCDA methods (Cinelli et al., 2020) and have developed a methodology to aid in the selection of appropriate MCDA methods based on the problem typology, preference model used, elicitation of decision maker preferences, and exploitation of the preferences (Cinelli et al., 2022). MCDA methods can be categorised into three main families; those which incorporate scoring functions, those which use binary relations, and those which use decision rules (Cinelli et al., 2022). For the interested reader an excellent overview of MCDA/MCDM methods is provided by (Wątróbski et al., 2019) as well as by (Figueira et al., 2016). Those who are interested in the history of MCDA/MCDM are directed to the work by (Köksalan et al., 2013).

MCDA is widely used across multiple fields. As of 2024 a Scopus search using the terms “MCDA” or “MCDM” returned 15,006 articles of which the majority are in the subject areas of Engineering (16.7 %), Computer Science (15.4 %), Environmental Science (10.3 %), Mathematics (8.8 %), and Business, Management and Accounting (8 %). The top three journals in terms of article number were Sustainability (483), Expert Systems with Applications (307), and Journal of Cleaner Production (230). A 2023 Google Scholar search using the term “MCDM” returned 131,000 results and using “MCDA” returned 62,100 results. The use of MCDA techniques is widespread across multiple disciplines including; sustainable energy (Wang et al., 2009), nature conservation (Adem Esmail & Geneletti, 2018), smartphone selection (Irvanizam et al., 2018), sustainable development (Kandakoglu et al., 2019), asset management (Syed & Lawryshyn, 2020), the distribution of social aid (Irvanizam et al., 2022), sustainability research (Sotoudeh-Anvari, 2023), and many more.

The broad range of fields within which MCDA can be found is a result of the ability of these methods to consider the performance of alternatives with respect to several criteria and coalesce these different performances into ranking or rating of the alternatives from “best” to “worst”. One of the most commonly used MCDA aggregation methods in literature is the weighted sum model (WSM) based on the L_1 Minkowski norm in the ranking of alternatives from best to worst (Greco et al., 2008).

1.2. Research problem

A key part of MCDA is sensitivity analysis, wherein changes in the results obtained are assessed for perturbations in the input data. This is particularly important for subjective values assigned to criteria weights (representative of the relative importance of criteria in the analysis or the level of trade-off allowable between criteria in compensatory methods) provided by decision makers. This paper focuses on sensitivity analysis with regards to criteria weights and includes a short review of sensitivity analysis methods developed in the literature (Section 3) that highlights the current state of the art and limitations therein. Following this, a novel and precise method to conduct sensitivity analysis with a commonly used MCDA aggregation method is developed and applied to an example problem.

1.3. Contributions of this work

This paper presents several novel and meaningful contributions in the field of sensitivity analysis for MCDM/MCDA approaches as follows:

- It expands the work first reported in (Triantaphyllou and Sánchez, 1997) by considering cost and benefit criteria and a ranking of the alternatives based on distance measures from the Nadir or Utopia alternatives.
- It derives weight stability intervals using closed form solutions instead of enumeration or simulation-based approaches which can be time consuming and imprecise.
- It considers weight stability intervals (WSIs) with the new closed form formulas for the determination of the most critical criterion/criteria. This is based on a Pareto dominance approach. This offers a holistic and effective way to consider the sensitivity analysis issue in a novel manner.

The remainder of this article is organised as follows: Section 2 introduces sensitivity analysis in conjunction with MCDA, Section 3 outlines the aim and objectives of this work. Section 4 details the methodology developed to perform a systematic sensitivity analysis based on weight stability intervals and applies this method to an example problem of renewable energy (biogas) production which is complex in terms of conflicting performance across multiple criteria and is also an important issue in its own right. Section 5 presents and discusses the results of the method applied to the example problem. Section 6 concludes this work.

2. Sensitivity of MCDA results to criteria weights: A brief review

If criteria weights are derived from subjective decision maker(s) input(s) there may be some uncertainty regarding their precise value. A key consideration in any MCDA is the sensitivity of the results obtained to alterations of the criteria weights used. The concept of sensitivity analysis is a key approach in assessing the robustness of the MCDA results obtained (Więckowski & Salabun, 2023). A commonly used method to identify the most critical or most sensitive criteria weight relies on identifying the smallest increase or decrease in criteria weights that can alter the ranking of the alternatives considered by a given MCDA method. This can be followed by a focused reanalysis of the critical criteria weights and can improve the decision-making process (Triantaphyllou, 2000) and (Demir et al., 2024).

A review, focused on the application of MCDA techniques to sustainability problems, estimates that only 30 % of the studies conducted some form of sensitivity analysis (Diaz-Balteiro et al., 2017). Approximately 44 % of 153 MCDA studies reviewed in the field of waste to energy management were found to contain some form of sensitivity analysis (Vlachokostas et al., 2021). A review of 343 MCDA articles related to sustainable development found that only 21 % of the studies applied sensitivity analysis to uncertain parameters such as criteria weights despite the recognised importance of sensitivity analysis (Kandakoglu et al., 2019). In a recent review of 250 MCDA articles that included sensitivity analysis, (Więckowski & Salabun, 2023) found that the manual alteration of criteria weights, or the use of several different scenarios of criteria weights was used in 77 % of the reviewed articles. A bibliometric analysis of 14,529 MCDA articles by (Demir et al., 2024) found that 9.5 % of publications used sensitivity analysis to test the robustness of the results obtained and that the manual alteration of criteria weights, or the use of several different scenarios of criteria weights was a commonly used sensitivity analysis method.

A form of sensitivity analysis commonly applied in the MCDA literature is the manual alteration of criteria weights by a set percentage of their initial values (e.g., +/- 5 %, +/-15 %, or +/-50 %) (Sahabuddin & Khan, 2021), the use of different sets of criteria weights (Dewalkar & Shastri, 2022) in a form of scenario analysis to assess whether the

ranking of the alternatives will change, or the process of searching for the maximum and minimum allowable alterations to individual criteria weights to determine a weight stability interval (Opricovic & Tzeng, 2007). The COMprehensive Sensitivity Analysis Method (COMSAM) is a further evolution beyond the iterative alteration of single criteria weights and performance under given criteria in which multiple alterations are made simultaneously allowing for the identification of intervariable dependencies (Więckowski & Sałabun, 2024). Recent software tools such as PySensMCDA (Paradowski et al., 2024) have expanded the ability to complete sensitivity analysis with multiple different MCDA methods.

Regarding the WSM as an aggregation step specifically, (Butler et al., 1997) used Monte Carlo simulations to assess the impact of altering criteria weights on the final ranking of alternatives obtained for coal power plant locations. Incremental alteration of criteria weights from -20% to $+20\%$ of their original value was used in a form of sensitivity analysis with the AHP method using the WSM as an aggregation step by (Chen et al., 2010) in ArcGIS for land suitability modelling. An enumeration approach where individual criteria weights were altered incrementally was completed by (Karande et al., 2016) in combination with the WSM to determine the individual weight stability intervals when ranking industrial robots. Manual alteration of criteria weights by $\pm 5\%$ and $\pm 50\%$, along with the use of 5 different scenarios of criteria weights was implemented as a form of sensitivity analysis by (Lee & Chang, 2018) when using the WSM to rank renewable energy options in Taiwan. An iterative alteration of criteria weights to determine individual weight stability intervals for the WSM was also completed by (Chauvy et al., 2020) to compare carbon dioxide utilisation technologies, and the use of 5 different scenarios of criteria weights was used by (Effatpanah et al., 2022) as a form of sensitivity analysis with the WSM when evaluating clean energy technologies in China.

A number of methods for completing a systematic sensitivity analysis have been developed in MCDA literature. An explicit closed form method developed by (Mareschal, 1988) enables the calculation of individual weight stability intervals (the range of values for individual weights for which the ranking of alternatives will not change) for the preference ranking organization method for enrichment of evaluations (PROMETHEE) method. In (Doan & De Smet, 2018) the authors used a mixed integer linear programming (MILP) model which is capable of determining the WSI for multiple criteria weights simultaneously. In (Huang et al., 2020) the authors propose an MILP model to derive the WSIs for multiple criteria weights, considering multiple decision makers. Additionally, (Triantaphyllou & Sánchez, 1997) developed a method to identify the most sensitive criteria weight when using the WSM as an aggregation step based on the smallest allowable change in criteria weights without changing the current ranking, or the top ranked alternative. A method to determine weight stability intervals to satisfy an a-priori specified preference order of alternatives was developed by (Wei et al., 2000) for use with the WSM as an aggregation step in the absence of a predefined set of weights from a decision maker. This was implemented in a Generic Multi-Attribute Analysis System software package based on additive multi-attribute utility by (Jiménez et al., 2006). Ordinal regression (Greco et al., 2008), robust ordinal regression (Corrente et al., 2013), and stochastic ordinal regression (Kadziński & Michalski, 2016) aim to obtain a full ranking of alternatives based on an a-priori specified preference/ranking of a subset of alternatives by a decision maker without eliciting criteria weights from the decision maker and whilst considering multiple possible value functions. A method to determine an interval of values for weights used in the ELECTRE TRI sorting method was developed and used by (Dias & Mousseau, 2003) and (Damart et al., 2007). Work by (Li et al., 2013) and (Alinezhad et al., 2009) developed methods to determine whether a given alteration to the value of a single criterion weight alters the ranking of the alternatives according to the TOPSIS method.

The only prior work dedicated to the WSM is by (Triantaphyllou &

Sánchez, 1997) and (Triantaphyllou, 2000) which identified the most sensitive criterion weight based on the minimum allowable increase or decrease to maintain the full ranking of alternatives, or to maintain the top ranked alternative. The calculation of weight stability intervals was not explicitly outlined by (Triantaphyllou & Sánchez, 1997) and (Triantaphyllou, 2000) but is possible using their method. However, calculation of weight stability intervals to maintain a partial ranking (for example to maintain the top 3 ranked alternatives) was not addressed. Furthermore, it may be possible for a criterion weight to have an asymmetrical weight stability interval (where the allowable increase and decrease in a criterion weight differ). In this event it is possible that one criterion may be the most sensitive when considering the allowable decrease in weight, whilst another criterion may be the most sensitive when considering the allowable increase in weight. Consideration of both the allowable increase and decrease in criteria weights (i.e. the extents of the weight stability intervals) has not been used previously to identify the most sensitive criterion weight.

Several knowledge gaps currently exist in the literature. Firstly, the use of an explicit closed form solution that does not depend on computationally intensive enumeration or mixed integer linear optimisation models to precisely identify the exact maximum and minimum allowable value for each criterion weight (the weight stability interval) to avoid a full or partial change in ranking has not been conducted for the WSM as an aggregation step in MCDA. Secondly, assessing the sensitivity of the result obtained from the WSM whilst considering both the allowable increase and decrease in criteria weights (the extreme points of the weight stability intervals) has not been addressed. The use of such a closed form analytical solution will be more accurate than the use of enumeration or simulation-based approaches which are widely used in the literature.

3. Objectives

The first objective of this work is to develop a method for the calculation of weight stability intervals for the WSM aggregation step with a closed form analytical solution that does not rely on enumeration. The second objective is to use these weight stability intervals to identify the most sensitive criterion weight whilst considering the allowable increase and decrease in criteria weights through a Pareto dominance-based approach.

4. Materials and methods

The following sections outline: 1) Common notation and concepts used within MCDA methods to aid in the clarity of this paper. 2) The development of a methodology for determining weight stability intervals for the widely used WSM aggregation step (which assumes full compensation between criteria) using closed form solutions. The stability interval for each criterion weight is expressed in terms of both the original non-normalised weights (W_j), and in terms of the normalised weights (w_j). This work expands upon that of (Triantaphyllou & Sánchez, 1997) by considering both cost and benefit criteria, and the measurement of “distance” from the Nadir or Utopia solutions, respectively. 3) Three cases of sensitivity analysis including a novel approach based on Pareto dominance. 4) Application of the methodology to an example problem.

4.1. Notation and concepts: Alternative performance, criteria weighting, and distance-based methods

Within MCDA methods, the performance of each alternative A_i ($i = 1, \dots, n$) with respect to a given criterion (c_j) can be expressed as p_{ij} in native units of the criteria in a performance matrix $P = (p_{ij}) \in \mathbb{R}^{n \times k}$. In this work it is assumed that the elements of P belong to ratio scales as they contain the most information as compared to ordinal or interval

scales. For each criterion $c_j (j = 1, \dots, k)$, the subjective relative importance $W_j \geq 0$ (commonly known as the criteria weights) of the criterion to the decision maker(s), or the allowable level of trade-off between criteria in MCDA methods that allow for full or partial compensation, can be accounted for in the comparison of alternatives. Criteria weights may be used as trade-offs in MCDA aggregation methods which allow for full and partial compensation (such as the WSM which is the focus of this work), or the weights may be used as importance coefficients in MCDA methods based on outranking (such as PROMETHEE) (Cinelli et al., 2020).

Various methods are available to estimate the value of these subjective weights including but not limited to; the AHP (Saaty, 1977), the weight elicitation stage of the simple multi attribute rating technique (SMART) using direct ratings of criteria importance (Edwards, 1977), modification of the weight elicitation stage of SMART incorporating swing weighting (SMARTS), further modification of the weight elicitation stage of SMART exploiting ranks (SMARTER) (Edwards & Barron, 1994), and the best-worst method (BWM) (Rezaei, 2015). The normalised weight, $0 \leq w_j \leq 1$, for c_j can then be expressed as $w_j = W_j / \sum_{l=1}^k W_l$. This work will not focus on how criteria weights are derived, rather it will focus on a novel method of sensitivity analysis as outlined in Section 2.

Typically, criteria units and scales differ, and normalisation may be required to express the performance values obtained by each alternative in the same units and on the same scale. Compensatory distance based MCDA aggregation methods (such as the widely used WSM) which can be used for the ranking of alternatives, with respect to c_j , based on a scoring method rely on the concept of “distance” from the utopia solution, $Utopia_j$, or the nadir solution, $Nadir_j$ (Yu, 1973) and (Zeleny, 1974), where;

$$Utopia_j = \begin{cases} \max(p_{1j}, \dots, p_{nj}), & \text{if } c_j \text{ is a benefit criterion to be maximised} \\ \min(p_{1j}, \dots, p_{nj}), & \text{if } c_j \text{ is a cost criterion to be minimised} \end{cases}$$

$$Nadir_j = \begin{cases} \min(p_{1j}, \dots, p_{nj}), & \text{if } c_j \text{ is a benefit criterion to be maximised} \\ \max(p_{1j}, \dots, p_{nj}), & \text{if } c_j \text{ is a cost criterion to be minimised} \end{cases}$$

We may use these reference points for the normalisation process. For each criterion c_j , the contents of P can then be normalised representing the deviation, x_{ij} , of A_i from the Nadir value (equation (1)), i.e.,

$$x_{ij}^{Nadir} = |Nadir_j - p_{ij}| / |Nadir_j - Utopia_j| \quad (1)$$

and in a similar fashion (equation (2)),

$$x_{ij}^{Utopia} = |Utopia_j - p_{ij}| / |Nadir_j - Utopia_j| \quad (2)$$

$Nadir_j$ can never be equal to $Utopia_j$, otherwise c_j would not contribute anything to the problem (it would be a redundant criterion). The concept of “distance”, d_i , of an alternative A_i from the Utopia solution or the Nadir solution was initially proposed by (Yu, 1973) and (Zeleny, 1974) based on the family of L_p metrics (equation (3)):

$$d_i = L_p(A_i) = \left(\sum_{j=1}^k w_j^p x_{ij}^p \right)^{\frac{1}{p}} \quad (3)$$

If d_i being considered is the distance from the Utopia (Nadir) solution, then the alternative with the minimum (maximum) d_i is the most preferable. It is possible for the most preferable alternative based on the minimum distance from Utopia to be different to the most preferable alternative based on the maximum distance from Nadir.

The parameter value p in L_p places a differing degree of emphasis on the weighted normalised performance values of the alternatives and in general $p \in \{1, 2, \infty\}$. When $p = 1$ the distance is equivalent to the

“Manhattan distance” or L_1 Minkowski norm, which is the longest distance in a geometric sense, when $p = 2$ the distance is equivalent to the “Euclidian distance” which is the shortest distance in a geometric sense, and when $p = \infty$ the distance is equivalent to the “Tchebycheff distance” which is the shortest in a numerical sense as the value of L_p decreases with an increase in p (Yoon, 1987). The choice of the distance metric is of paramount importance since it is a proxy measure for human preferences (Romero & Rehman, 2003). When $p = 1$ emphasis is placed on seeking the largest achievement, while when $p = \infty$ the analyst prioritizes giving up large achievement for balanced solutions (Ballestero, 2007).

When $p = 1$, equation (3) reduces to a weighted linear combination of the normalised values (equation (4)) as all exponents in equation (3) equal one:

$$d_i = L_1(A_i) = \sum_{j=1}^k w_j x_{ij} \quad (4)$$

Equation (4) is equivalent to the simple additive weighting (SAW), or weighted sum model (WSM) aggregation step within the MCDA method first proposed by (Fishburn, 1967) and is the most widely used MCDA aggregation method across multiple fields owing to its ease of interpretation and simplicity compared to other MCDA aggregation methods. The WSM method was found to be the most widely used MCDA aggregation method in numerous literature reviews in disciplines such as sustainable energy (Wang et al., 2009), fishery management (Andalecio, 2009), nature conservation (Adem Esmail & Geneletti, 2018), sustainable development (Kandakoglu et al., 2019), and indeed a review of MCDA methods themselves (Triantaphyllou & Mann, 1989) and (Triantaphyllou, 2000). It is for this reason that the WSM method is the focus of this work.

4.2. WSM weight stability interval calculation

For the purpose of this paper the “distance” from Nadir will be used. As such, alternatives with a greater “distance” from Nadir, as given in (3), are preferable. The methodology developed herein can also be used when the distance to Utopia is considered. Of interest, when considering the WSM method the use of “distance” from Nadir or “distance” from Utopia does not alter the ranking of the alternatives.

Theorem 1. If two alternatives A_K and A_L are ranked as $A_K > A_L$ under the WSM method when the Nadir alternative is used for reference, then these two alternatives will be ranked in exactly the same way when the Utopia alternative is considered, and vice-versa.

This theorem can be proved as follows:

Consider two alternatives, for simplicity in the notation, we denote them as alternatives A_1 and A_2 , in terms of criterion c_j . This criterion may be a benefit or a cost criterion. Next, we compare them in terms of the distance measure each has from the Nadir alternative.

Let us consider the normalised values x_{1j}^{Nadir} and x_{2j}^{Nadir} defined by equation (1) and let us also assume that criterion c_j is a benefit criterion (the higher the value, the better it is). These two normalised values are defined as follows (equation (5)):

$$x_{ij}^{Nadir} = \frac{|Nadir_j - p_{ij}|}{|Nadir_j - Utopia_j|}, \quad \text{for } i = 1, 2 \quad (5)$$

Since criterion c_j is assumed to be a benefit criterion, we always have $Nadir_j \leq p_{ij}$, for any applicable value of i . Therefore, we will have:

$$|Nadir_j - p_{ij}| = (p_{ij} - Nadir_j).$$

From the last result, equation (5) becomes equation (6):

$$x_{i,j}^{Nadir} = \frac{|Nadir_j - p_{1i,j}|}{|Nadir_j - Utopia_j|} = \frac{(p_{1i,j} - Nadir_j)}{|Nadir_j - Utopia_j|}, \text{ for } i = 1, 2. \quad (6)$$

Next, define as $D_{1,2}^{Nadir,j}$ the difference of the distance measures from the Nadir alternative for the two alternatives A_1 and A_2 but only in terms of criterion c_j . From equation (6), the following derivations can easily become evident yielding equation (7):

$$\begin{aligned} D_{1,2}^{Nadir,j} &= (w_j x_{1,j}^{Nadir} - w_j x_{2,j}^{Nadir}) = w_j (x_{1,j}^{Nadir} - x_{2,j}^{Nadir}), \text{ or:} \\ D_{1,2}^{Nadir,j} &= w_j \frac{((p_{1,j} - Nadir_j) - (p_{2,j} - Nadir_j))}{|Nadir_j - Utopia_j|} \text{ or:} \\ D_{1,2}^{Nadir,j} &= \frac{w_j (p_{1,j} - p_{2,j})}{|Nadir_j - Utopia_j|} \end{aligned} \quad (7)$$

Now the previous derivations are considered, but in terms of the Utopia alternative. Thus, we have the two normalized values $x_{1,j}^{Utopia}$ and $x_{2,j}^{Utopia}$ defined in equation (8):

$$x_{i,j}^{Utopia} = \frac{Utopia_j - p_{i,j}}{|Nadir_j - Utopia_j|}, \text{ for } i = 1, 2. \quad (8)$$

The last expression is valid because since criterion c_j is a benefit criterion, $Utopia_j \geq p_{i,j}$, for any applicable value of i . Let the quantity $D_{1,2}^{Utopia,j}$ denote the difference of the distance measures from the Utopia alternative for the two alternatives A_1 and A_2 but only in terms of criterion c_j . When this entity is used, equation (8) leads to the following result in a manner similar to the previous derivations (equation (9)):

$$D_{1,2}^{Utopia,j} = \frac{w_j (p_{2,j} - p_{1,j})}{|Nadir_j - Utopia_j|} \quad (9)$$

When equations (7) and (9) are compared, the following (equation (10)) is evident, for each criterion c_j (assumed to be a benefit criterion):

$$D_{1,2}^{Nadir,j} = -D_{1,2}^{Utopia,j} \quad (10)$$

When cost criteria are considered, then a similar result as equation (10) is derived. Let the quantities $D_{1,2}^{Nadir}$ and $D_{1,2}^{Utopia}$ denote the difference in distances of the two alternatives A_1 and A_2 when the Nadir and Utopia alternatives are considered as reference points, respectively. Then, given equation (3), the following derivations can be obtained:

$$D_{1,2}^{Nadir} = \sum_j^k D_{1,2}^{Nadir,j}$$

$$D_{1,2}^{Utopia} = \sum_j^k D_{1,2}^{Utopia,j}$$

When the last result is combined with equation (10), equation (11) is derived:

$$D_{1,2}^{Nadir} = -D_{1,2}^{Utopia} \quad (11)$$

In other words, if alternative A_1 is closer to the Nadir alternative than alternative A_2 , then alternative A_1 will be further away from the Utopia alternative than alternative A_2 , and vice-versa. The above analysis completes the proof of Theorem 1.

Therefore, if we rank a set of alternatives using the WSM aggregation method, it does not matter whether we use the Nadir or the Utopia alternative as a reference point, the ranking will remain the same.

The L_1 distance from Nadir (as per the WSM), for alternative A_i , is then $d_i^{Nadir} = L_1(A_i) = \sum_{j=1}^k w_j x_{ij}^{Nadir}$. The goal is to identify the alternative with the largest value d_i^{Nadir} .

An alternative, A_E , performs better than another alternative, A_F , provided that $d_E^{Nadir} > d_F^{Nadir}$. In assessing the alteration α_j , which can be

applied to the normalised weight w_j to ensure that A_E always performs better than A_F , the updated weight w_j^* can be written as per equation (12);

$$w_j^* = (1 + \alpha_j) w_j \quad (12)$$

where, for w_j^* to remain non-negative, $\alpha_j \geq -1$.

In order for the relative weight ratios of the remaining weights to remain unchanged their normalised weights must be altered. The modified value $w_{i \neq j}^*$ can be found according to (13) and (14) adapted from (Mareschal, 1988);

$$\beta_j = \frac{1 - (1 + \alpha_j) w_j}{1 - w_j} \quad (13)$$

$$w_i^* = \beta_j w_i \quad (14)$$

where, for w_i^* to remain positive $\alpha_j < (1 - w_j) / w_j$.

Following an alteration to weight w_j the L_1 distance from Nadir of A_E , denoted as d_E^{Nadir*} , can be expressed as per equation (15). Note that x_{Ej} in all subsequent equations is equivalent to x_{Ej}^{Nadir} ;

$$d_E^{Nadir*} = \frac{((1 + \alpha_j) \cdot (1 - w_j) \cdot w_j \cdot x_{Ej} + (1 - w_j + \alpha_j w_j) \cdot (d_E^{Nadir} - w_j \cdot x_{Ej}))}{1 - w_j} \quad (15)$$

For alternative A_F to perform the same as alternative A_E , following an alteration to weight w_j , $d_E^{Nadir*} = d_F^{Nadir*}$.

The value of α_j which ensures that the above expression is valid can be found as per equation (16);

$$\alpha_j = \frac{(1 - w_j) (d_F^{Nadir} - d_E^{Nadir})}{w_j (x_{Ej} - x_{Fj} - d_E^{Nadir} + d_F^{Nadir})} \quad (16)$$

Based on equation (16) the values of α_j^+ and α_j^- (the positive/negative deviation that can be applied to w_j) for alternative A_E to perform better than alternative A_F can be determined as per equation (17) and equation (18);

$$\alpha_j^- = \begin{cases} \max(-1, \alpha_j), & \text{if } \alpha_j \leq 0 \\ -1, & \text{if } \alpha_j > 0 \end{cases} \quad (17)$$

$$\alpha_j^+ = \begin{cases} \min\left(\frac{1 - w_j}{w_j}, \alpha_j\right), & \text{if } \alpha_j \geq 0 \\ \frac{1 - w_j}{w_j}, & \text{if } \alpha_j < 0 \end{cases} \quad (18)$$

This process provides the normalised weight stability interval (wsi) for the normalised weight w_j of criterion c_j , namely $w_{sj} = [w_j(1 + \alpha_j^-), w_j(1 + \alpha_j^+)] = [w_{jl}, w_{ju}]$. This ensures that alternative A_E performs better than alternative A_F for an alteration to a single criterion weight, w_j , within $[w_{jl}, w_{ju}]$. The reader is reminded that as the sum of the normalised weights w_j must equal unity, the values of the remaining weights $w_{z|z \neq j}$ must be proportionally adjusted.

Calculation of the weight stability interval for the original unnormalised weight (W_j) prior to normalisation (when the criteria weights do not sum to 1) for criterion c_j can be found by converting the values of w_{jl} and w_{ju} to W_{jl} and W_{ju} as per equation (19);

$$W_{ju} = \frac{(1 + \alpha_j^+) \cdot W_j \cdot (\sum W - W_j)}{\sum W - (1 + \alpha_j^+) \cdot W_j} \quad (19)$$

The same procedure can be applied to calculate W_{jL} using α_j^- in place of α_j^+ , and yields the weight stability interval $WSI_j = [W_{jL}, W_{jU}]$ of the original un-normalised weight W_j for criterion c_j . The reader is reminded that a lowercase $wsij$ corresponds to the normalised weight w_j , while an uppercase WSI_j corresponds to the original un-normalised weight W_j . Following an alteration to the un-normalised weight W_j within $[W_{jL}, W_{jU}]$ the weights must then be normalised to ensure that they sum to 1. The

calculation procedure to determine $wsij$ and WSI_j when considering two alternatives A_E and A_F is outlined in Fig. 1.

From the n alternatives being assessed, the weight stability interval for w_j to ensure that the order of preference of the top q alternatives ($q \leq n$) is maintained can be identified by determining α_{jrs} in a sequential pairwise manner for the r^{th} and s^{th} best alternatives (the r^{th} alternative has a better rank than the s^{th} alternative). For example, if the

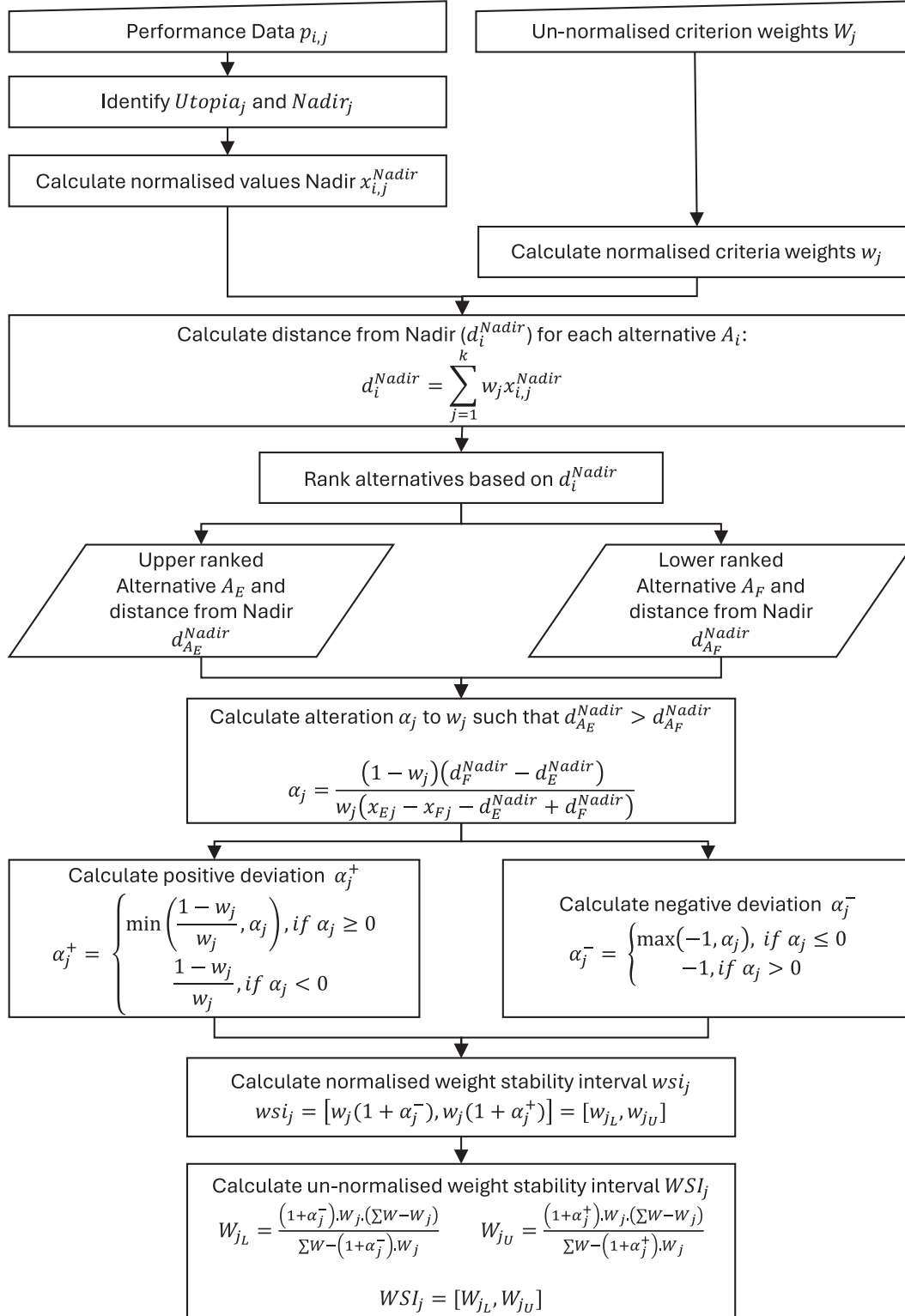


Fig. 1. Procedure to calculate weight stability intervals for normalised criteria weights ($wsij$) and un-normalised criteria weights (WSI_j), respectively.

weight stability interval for w_j is to be calculated to ensure that the order of preference for the top q alternatives (assuming that $d_1^{Nadir} > d_2^{Nadir} > d_3^{Nadir} > \dots d_q^{Nadir} > \dots d_n^{Nadir}$) remains unchanged the procedure outlined below is completed;

calculate $\alpha_{j1,2}$
 calculate $\alpha_{j2,3}$
 calculate $\alpha_{j3,4}$
 calculate $\alpha_{j4,5}$
 calculate $\alpha_{j5,6}$
 $\vdots$
 calculate $\alpha_{jq-1,q}$
 calculate $\alpha_{jq,q+1}$
 $\vdots$
 calculate α_{jn}

The reduction in w_j will be dictated by the minimum allowable absolute reduction as per equation (20). If all of the α_j are less than zero then the allowable reduction in w_j is; the maximum value of α_j which is less than zero but greater than -1 , or, -1 if all the values of α_j are less than -1 .

$$\alpha_j^- = \begin{cases} \max\{-1, \alpha_{j1,2}, \alpha_{j2,3}, \dots, \alpha_{jq-1,q}, \alpha_{jq,q+1}, \alpha_{jq,q+2}, \dots, \alpha_{jn}\}, & \alpha_{jrs} \leq 0 \\ -1, & \alpha_{jrs} > 0 \end{cases} \quad (20)$$

The increase in w_j will be dictated by the maximum allowable absolute increase as per equation (21). If all of the α_j are less than zero then the allowable increase in w_j is $\frac{1-w_j}{w_j}$, or, the minimum of $\frac{1-w_j}{w_j}$ and all values of α_j which are greater than or equal to zero.

$$\alpha_j^+ = \begin{cases} \min\left\{\frac{1-w_j}{w_j}, \alpha_{j1,2}, \alpha_{j2,3}, \dots, \alpha_{jq-1,q}, \alpha_{jq,q+1}, \alpha_{jq,q+2}, \dots, \alpha_{jn}\right\}, & \alpha_{jrs} \geq 0 \\ \frac{1-w_j}{w_j}, & \alpha_{jrs} < 0 \end{cases} \quad (21)$$

Again, the weight stability interval for the normalised weight w_j can be expressed using equation (22).

$$wsi_j = [w_j(1 + \alpha_j^-), w_j(1 + \alpha_j^+)] = [w_{jl}, w_{ju}] \quad (22)$$

The values of α_j^- and α_j^+ determined in equation (20) and equation (21) are the minimum absolute change in w_j which can be achieved without altering the order of preference for the top q alternatives.

The same procedure outlined previously to calculate WSI_j for the un-normalised criterion weight W_j can then be followed. When q is equal to n the weight stability intervals calculated (wsi_j for the normalised weight or WSI_j for the un-normalised weight) correspond to the stability intervals required to maintain full stability as per (Mareschal, 1988). When q is less than n the stability intervals correspond to the maximum and minimum range of values for each criterion weight w_j to ensure the order of preference within the top q alternatives is maintained.

4.3. Sensitivity analysis

The most sensitive criterion may be determined in several ways. Prior work by (Triantaphyllou, 2000) identified two methods for identifying the most sensitive criterion. The first considers the minimum allowable absolute change (AC) in the criterion weight ($AC_w = \min([W_j - W_{jl}, W_{ju} - W_j])$, $j = 1$ to k for the un-normalised weights or $AC_w = \min([w_j - w_{jl}, w_{ju} - w_j])$, $j = 1$ to k for the normalised weights). The second considers the minimum allowable relative (percentage) change (RC) in the criterion weight ($RC_w = \min\left(\left[\frac{W_j - W_{jl}}{W_j} * 100, \frac{W_{ju} - W_j}{W_j} * 100\right]\right)$, $j = 1$ to k for the un-

normalised weights or $RC_w = \min\left(\left[\frac{w_j - w_{jl}}{w_j} * 100, \frac{w_{ju} - w_j}{w_j} * 100\right]\right)$, $j = 1$ to k for the normalised weights). Within this work, the minimum allowable relative (percentage) change in the criterion weight (RC_w for normalised or RC_w for un-normalised weights, respectively) based on the weight stability interval for the normalised weights (wsi_j) or un-normalised weights (WSI_j), will be used as one method of identifying the most sensitive criterion (Case 1).

In addition to this, the most sensitive criterion will also be identified as that with the smallest weight stability interval expressed as the absolute difference between the allowable percentage increase and percentage decrease in the criterion weight ($RC_{wsi} = \min\left(\left|\frac{w_{ju} - w_j}{w_j} - \frac{w_{jl} - w_j}{w_j}\right| * 100\right)$, $j = 1$ to k for the un-normalised weights or $RC_{wsi} = \min\left(\left|\frac{w_{ju} - w_j}{w_j} - \frac{w_{jl} - w_j}{w_j}\right| * 100\right)$, $j = 1$ to k for the normalised weights) (Case 2).

Finally, Pareto dominance (considering minimisation of both the allowable absolute relative increase ($RI_{w_j} = \left|\frac{w_{ju} - w_j}{w_j} * 100\right|$ for un-normalised and $RI_{w_j} = \left|\frac{w_{ju} - w_j}{w_j} * 100\right|$ for normalised weights, respectively) and relative decrease ($RD_{w_j} = \left|\frac{w_j - w_{jl}}{w_j} * 100\right|$ for un-normalised and $RD_{w_j} = \left|\frac{w_j - w_{jl}}{w_j} * 100\right|$ for normalised weights, respectively) will be used to identify the most sensitive criterion weight(s) in a bi-objective problem (Case 3) as follows:

Criterion c_θ is dominated by $criterionc_\Omega$ if:

- $RI_{w_\Omega} < RI_{w_\theta}$ and $RD_{w_\Omega} < RD_{w_\theta}$ for un-normalised weights.
- $RI_{w_\Omega} < RI_{w_\theta}$ and $RD_{w_\Omega} < RD_{w_\theta}$ for normalised weights.
- All 3 cases are illustrated in the example problem outlined in Section 4.4.

4.4. Application to an example problem

The method developed in the prior section is applied to an example problem consisting of the selection of the “best” blend of feedstock material to use in an anaerobic digester (a device that can convert biodegradable material into a renewable gaseous fuel called “biogas”, and a residual effluent called “digestate”). A total of 27 alternatives are assessed and their performance with regards to 5 criteria; GHG emission saving (tCO₂e/q) a benefit criterion, digester volume (m³) a cost criterion, mass of digestate produced (t_{wwt}) a cost criterion, total solids content of feedstock (%_{wwt}) a cost criterion, and the energy density of the plant (MWh/m³) a benefit criterion, was quantified (see Table 1). Each of the 5 criteria are assumed to have an equal weight of importance and thus are assigned a weight of importance (W_j) of 0.2, see Table 2. In this example as the original weights of importance (W_j) already sum to 1, the normalised weight of importance of each criterion (w_j) is also 0.2. The weight stability intervals for the original weights (WSI) and for the normalised weights (wsi) are calculated to maintain the order of preference for all alternatives assessed, as well as for the top 5 alternatives.

The order of preference will remain unchanged if one criterion weight at a time is altered to a value within its given stability interval, and the remaining criteria weights are re-scaled accordingly. Calculation of the weight stability intervals is achieved using GNU Octave. To aid interpretation of the methodology, a step-by-step calculation is provided in Appendix A (Supplementary Data) for the first criterion (c_1) Scope 1 GHG Emission Saving.

Table 1
Alternative data.

Alternative	Scope 1 GHG Emission Saving	Digester Volume	Digestate Mass	Feedstock total solids	Energy Density of AD plant
	tCO ₂ eq/a	m ³	twwt/a	%wwt	MWh/m ³
A ₁	1,458.52	2,787.97	48,573.19	8.79	1.84
A ₂	895.92	1,858.64	35,811.80	7.92	1.69
A ₃	631.96	1,548.87	31,612.78	7.47	1.43
A ₄	2,237.08	3,717.29	51,885.07	10.88	2.11
A ₅	1,332.97	2,323.30	37,433.41	9.42	2.01
A ₆	1,031.01	1,858.64	32,616.71	8.64	1.95
A ₇	2,681.55	4,644.92	55,415.44	12.69	2.03
A ₈	1,908.31	2,787.97	38,956.05	10.78	2.40
A ₉	1,174.97	2,168.42	33,803.34	9.72	1.90
A ₁₀	900.99	1,904.49	36,281.20	8.12	1.66
A ₁₁	1,194.71	2,323.30	46,827.47	7.62	1.81
A ₁₂	885.91	1,858.64	38,917.32	7.32	1.67
A ₁₃	1,174.27	2,380.61	38,067.81	9.63	1.73
A ₁₄	1,433.90	2,787.97	48,590.82	8.79	1.81
A ₁₅	1,180.25	2,168.42	39,996.25	8.27	1.91
A ₁₆	1,478.22	2,856.74	39,832.52	10.98	1.82
A ₁₇	2,026.14	3,252.63	50,101.41	9.87	2.19
A ₁₈	1,698.82	2,478.19	40,914.66	9.17	2.41
A ₁₉	1,006.07	1,587.08	31,567.61	7.77	2.23
A ₂₀	1,264.33	1,904.49	39,337.87	7.43	2.33
A ₂₁	1,153.90	2,168.42	46,211.83	7.21	1.87
A ₂₂	552.31	1,904.49	33,214.03	8.94	1.02
A ₂₃	731.87	2,221.91	41,040.68	8.39	1.16
A ₂₄	906.22	2,478.19	47,678.92	8.02	1.28
A ₂₅	793.24	2,221.91	34,363.04	10.03	1.25
A ₂₆	3,004.35	2,539.32	40,734.97	9.30	4.15
A ₂₇	6,294.61	4,668.52	53,398.87	12.63	4.73

Table 2
Criteria weighting.

	Scope 1 GHG Emission Saving	Digester Volume	Digestate Mass	Feedstock total solids	Energy Density of AD plant
Un-normalised weight	0.2	0.2	0.2	0.2	0.2
Normalised weight	0.2	0.2	0.2	0.2	0.2
Goal	Maximise	Minimise	Minimise	Minimise	Maximise

5. Results and discussion

5.1. Weight stability intervals

The results of the MCDA using the WSM (L_1) aggregation method are presented in Table 3 and indicate that the top 5 alternatives in order of decreasing preference are; $A_{19} > A_{26} > A_3 > A_{20} > A_6$.

The weight stability intervals for each criterion within which the order of ranking for all 27 alternatives is maintained are provided in Table 4. The weight stability intervals which ensure the order of ranking for the top 5 alternatives is maintained are provided in Table 5.

A graphical representation of the allowable percentage increase and decrease of the un-normalised criteria weights for the WSM (L_1) aggregation method to maintain the order of ranking for all 27 alternatives and to maintain the order of ranking for the top 5 alternatives is provided in Figs. 2 and 3, respectively. Alteration of a single criterion weight within the limits in Fig. 2 will not change the rankings obtained. The width of an individual bar (be it an increase or decrease) may be used to determine the most sensitive criterion according to Case 1 (based on the minimum allowable percentage increase or decrease, RC_W for un-normalised weights). The combined width of the bars in Fig. 2 are equivalent to the measure of sensitivity applied in Case 2 (RC_{WSI} for un-normalised weights), where the most sensitive criterion has the smallest combined width.

Fig. 3 provides an alternative visualisation of the results where criteria weights with a narrower weight stability interval from Case 2 are close to the origin. Fig. 3 may be used to identify the most sensitive criterion weight through a Pareto dominance approach based on RI_W .

Table 3

Alternative ranking based on the WSM (L_1) aggregation method.

Alternatives	$L1 : d^{Nadir}$	Ranking Preference
A ₁	0.3960	22
A ₂	0.5667	6
A ₃	0.6150	3
A ₄	0.2741	26
A ₅	0.5011	13
A ₆	0.5860	5
A ₇	0.1301	27
A ₈	0.4499	18
A ₉	0.5191	10
A ₁₀	0.5511	8
A ₁₁	0.4724	15
A ₁₂	0.5611	7
A ₁₃	0.4638	16
A ₁₄	0.3934	23
A ₁₅	0.5208	9
A ₁₆	0.3846	24
A ₁₇	0.3527	25
A ₁₈	0.5054	11
A ₁₉	0.6581	1
A ₂₀	0.5994	4
A ₂₁	0.5042	12
A ₂₂	0.5003	14
A ₂₃	0.4481	19
A ₂₄	0.4021	21
A ₂₅	0.4513	17
A ₂₆	0.6375	2
A ₂₇	0.4191	20

Table 4

Weight stability intervals to ensure order of preference of all 27 alternatives is maintained.

Weighted Sum Method (L_1)	C1 Scope 1 GHG Emission Saving	C2 Digester Volume	C3 Digestate Mass	C4 Feedstock Total Solids	C5 Energy Density of AD Plant ^{A, B, a, b}
Original weight	0.2	0.2	0.2	0.2	0.2
W_{ju}	0.2070	0.2060	0.2046	0.2031	0.2044
W_{jl}	0.1941	0.1926	0.1949	0.1936	0.1970
Allowable weight increase (%)	3.48	3.00	2.28	1.57	2.18
Allowable weight decrease (%)	-2.96	-3.72	-2.53	-3.22	-1.51
Increase – decrease (%)	6.44	6.72	4.80	4.79	3.69
Normalised weight	0.2	0.2	0.2	0.2	0.2
w_{ju}	0.2055	0.2048	0.2036	0.2025	0.2035
w_{jl}	0.1952	0.1940	0.1959	0.1948	0.1976
Allowable weight increase (%)	2.76	2.39	1.81	1.25	1.74
Allowable weight decrease (%)	-2.39	-3.00	-2.03	-2.60	-1.21
Increase – decrease (%)	5.15	5.39	3.84	3.85	2.95

A Most sensitive un-normalised weight – Case 1, minimum allowable percentage increase or decrease (RC_W).

B Most sensitive un-normalised weight – Case 2, minimum difference between percentage increase and decrease (RC_{WSI}).

a Most sensitive normalised weight – Case 1, minimum allowable percentage increase or decrease (RC_W).

b Most sensitive normalised weight – Case 2, minimum difference between percentage increase and decrease (RC_{WSI}).

and RD_{W_j} for un-normalised weights (Case 3). When seeking to maintain the full order of all 27 alternatives the Pareto dominant criteria weights (therefore the most sensitive) are C4 (Feedstock Total Solids) and C5 (Energy Density of AD Plant) based on the allowable relative increase (RI_{W_4} and RI_{W_5}) and allowable relative decrease (RI_{W_4} and RI_{W_5}). Depending on the relative importance of an increase or decrease in criterion weight to the decision maker(s) either C4 or C5 will be the most sensitive criterion. If the ranking of the top 5 alternatives is to be maintained there is only 1 Pareto dominant criterion, C5 (Energy Density of AD Plant), regardless of the relative importance of an increase or decrease in criteria weights.

A graphical interpretation of the un-normalised weight stability intervals (WSI_j) for each criterion to ensure that the order of preference for the top 5 alternatives is maintained is provided in Fig. 4. Fig. 4 was developed by manually altering the individual criterion weight under consideration in an iterative approach, the most common sensitivity analysis method used with the WSM aggregation method. The points at which the lines corresponding to $d^{Nadir}(L_1)$ for each alternative intersect indicate W_{jl} and W_{ju} , respectively. The graphical results correspond well to the results in Table 5.

Table 5

Weight stability intervals to ensure order of preference of the top 5 alternatives is maintained.

Weighted Sum Method (L_1)	C1 Scope 1 GHG Emission Saving	C2 Digester Volume	C3 Digestate Mass	C5 Feedstock Total Solids	C5 Energy Density of AD Plant ^{A, B, a, b}
Original weight	0.2	0.2	0.2	0.2	0.2
W_{ju}	0.2594	0.2708	0.2478	0.2673	0.2399
W_{jl}	0.1456	0.1323	0.1519	0.1390	0.1694
Allowable weight increase (%)	29.69	35.39	23.89	33.64	19.97
Allowable weight decrease (%)	-27.19	-33.85	-24.05	-30.49	-15.32
Increase – decrease (%)	56.89	69.24	47.94	64.14	35.29
Normalised weight	0.2	0.2	0.2	0.2	0.2
w_{ju}	0.2449	0.2529	0.2365	0.2504	0.2307
w_{jl}	0.1540	0.1419	0.1596	0.1480	0.1747
Allowable weight increase (%)	22.43	26.44	18.24	25.22	15.36
Allowable weight decrease (%)	-23.01	-29.05	-20.22	-25.98	-12.65
Increase – decrease (%)	45.43	55.49	38.46	51.20	28.01

A Most sensitive un-normalised weight – Case 1, minimum allowable percentage increase or decrease (RC_W).

B Most sensitive un-normalised weight – Case 2, minimum difference between percentage increase and decrease (RC_{WSI}).

a Most sensitive normalised weight – Case 1, minimum allowable percentage increase or decrease (RC_W).

b Most sensitive normalised weight – Case 2, minimum difference between percentage increase and decrease (RC_{WSI}).

5.2. Discussion

5.2.1. Weight stability intervals

The method developed within this work allows for an accurate and exact calculation of the weight stability intervals for the weight of each criterion when using the WSM method as an aggregation step within MCDA (based on the L_1 distance). The weight stability intervals obtained for the un-normalised criteria weights to maintain the full order of alternatives match the results obtained using the method outlined by (Triantaphyllou & Sánchez, 1997), results are provided in Appendix B. As outlined in the introduction, the majority of sensitivity analyses completed within the field of MCDA rely on; scenario analysis in which criteria weights are altered by a set amount, computationally intensive enumeration-based approaches where criteria weights are systematically altered, simulation-based approaches that use Monte Carlo methods, or mixed integer linear programming optimisation methods (MILP).

Of the prior methods, only those using an MILP approach can provide an exact calculation of the allowable increase or decrease for a given criterion weight (Doan and De Smet, 2018) and (Huang et al., 2020). However, these MILP models can be more computationally intensive than the explicit methods developed in this work and may also rely on

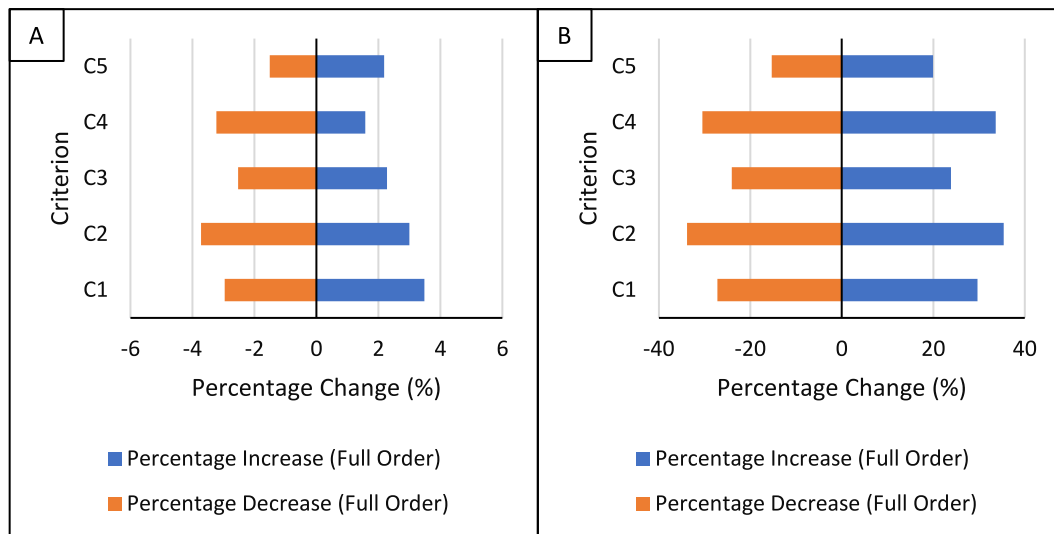


Fig. 2. Allowable percentage increase and decrease in un-normalised criteria weights to maintain the order of preference of alternatives. A: WSM Full order of 27 alternatives maintained, B: WSM Order of top 5 alternatives maintained. C1: Scope 1 GHG Emission Saving, C2: Digester Volume, C3: Digestate Mass, C4: Feedstock Total Solids, C5: Energy Density of AD Plant.

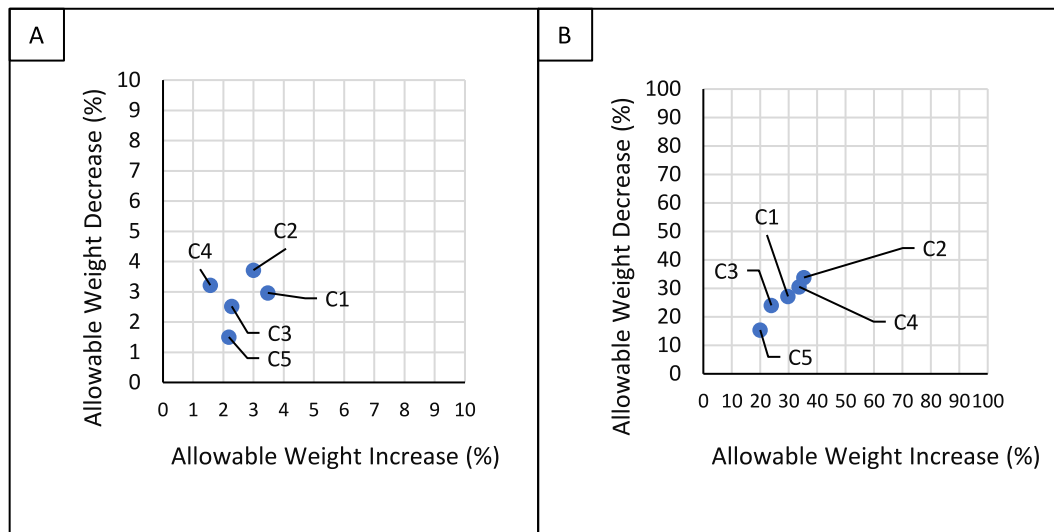


Fig. 3. Allowable percentage increase and decrease in un-normalised criteria weights to maintain the order of preference of alternatives. A: WSM Full order of 27 alternatives maintained, B: WSM Order of top 5 alternatives maintained.

commercial software presenting a barrier to some users. The weight stability intervals determined in this work are similar to those obtained by; (Mareschal, 1988) for use with the PROMETHEE method. However, this work is the first to present an exact method of determining the weight stability intervals for the WSM aggregation method to maintain the complete ranking of alternatives, or a subset of the top q ranked alternatives ($q \leq n$). This extends beyond the prior work by (Triantaphyllou & Sánchez, 1997) who proposed a method to identify the most sensitive criterion by calculating the minimum allowable absolute or relative change (increase or decrease) in the individual criteria weights to maintain the full ranking of alternatives. The weight stability interval method presented in this work allows for the exact calculation of both the allowable increase and decrease for each criterion weight to maintain the full ranking of alternatives, or, to maintain the ranking of the top q alternatives. As such, the method developed in this work improves upon the state-of-the-art methods of conducting sensitivity analysis for the WSM method.

5.2.2. Sensitivity case 1

The weight stability interval for Scope 1 GHG Emission Saving (C1) and Digester Volume (C2) indicates that these normalised weights can be altered by the greatest degree (+2.76 %/-2.39 % and +2.39 %/-3.00 %, respectively) without changing the ranking of all 27 alternatives (Fig. 2A). The original un-normalised weights of Digestate Mass (C3) and Feedstock Total Solids Content (C4) have a narrower range of permissible weight alteration (+2.28 %/-2.53 % and +1.57 %/-3.22 %, respectively) without changing the ranking of all 27 alternatives. The narrowest range of normalised weight alteration can be applied to the Energy Density of AD Plant (C5) (+2.18 %/-1.51 %). The narrow stability interval for all criteria indicates that the order of preference when all 27 alternatives are considered is very sensitive to alterations to the original weights.

The most sensitive criterion weight when aiming to preserve the ranking of all 27 alternatives based on the smallest allowable absolute percentage change in the original un-normalised weight (-1.51 %) or the normalised weight (-1.21 %) is the Energy Density of AD Plant (C5)

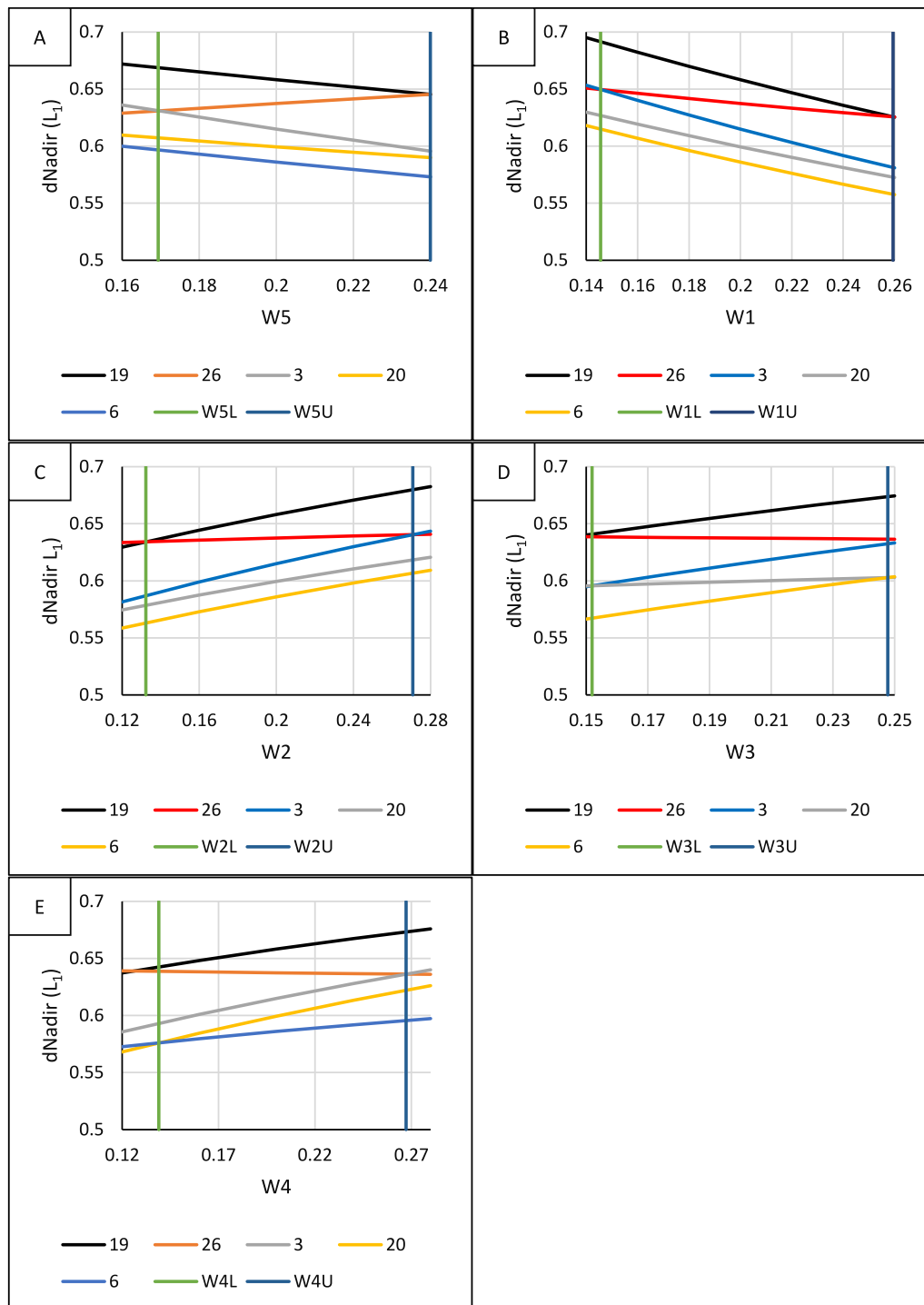


Fig. 4. Weighted Sum Method (L1): Weight stability intervals to maintain order of preference for top 5 alternatives. A: Scope 1 GHG Emission Saving (W1). B: Digester Volume (W2). C: Digestate Mass (W3). D: Feedstock Total Solids Content (W4). E: Energy Density of AD Plant (W5).

(Fig. 2A). These results agree with the results obtained when using the method developed by (Triantaphyllou & Sánchez, 1997) in Appendix B.

If the order of preference for the top 5 alternatives is to be maintained the weight stability intervals are at least as wide as those to maintain the order of preference for all 27 alternatives (Fig. 2B). The un-normalised weight associated with Digester Volume (C2) may be altered by the largest degree (+35.39 %/-33.85 %) indicating that this criterion is the least sensitive when the preference for the top 5 alternatives is to be maintained. Feedstock Total Solids Content (C4) (+33.64 %/-30.49 %) and Scope 1 GHG Emission Saving (C1) (+29.69 %/-27.19 %) have

the next narrowest range of permissible weight alteration (based on the un-normalised weights) indicating a greater sensitivity of the result to these criteria. Finally, Digestate Mass (C3) (+23.89 %/-24.05 %) and Energy Density of the AD Plant (C5) (19.97 %/-15.32 %) are the criteria with the narrowest range of allowable alterations in un-normalised criteria weight, indicating that the result of the WSM is most sensitive to the weights of these 2 criteria when seeking to maintain the order of preference for the top 5 alternatives.

The most sensitive criterion weight when aiming to preserve the ranking of the top 5 alternatives is Energy Density of the AD Plant (C5)

as it has the smallest allowable absolute percentage change based on the weight stability interval using the original weights (−15.32 %) and the normalised weights (−12.65 %), see Fig. 2B, again these results agree with those obtained using the methodology developed by (Triantaphyllou & Sánchez, 1997).

5.2.3. Sensitivity case 2

The Energy Density of the AD Plant (C5) is also the most sensitive criterion when considering the absolute difference between the allowable percentage increase and percentage decrease in the criterion weight (3.69 % for the un-normalised weight and 2.95 % for the normalised weight), when aiming to preserve the ranking of all 27 alternatives.

When aiming to preserve the ranking of the top 5 alternatives, the weight associated with the Energy Density of the AD Plant (C5) is once again the most sensitive (35.29 % for the un-normalised weight and 28.01 % for the normalised weight). However, this agreement between the 2 definitions of sensitivity (Case 1 and Case 2) may not always occur and could lead to an interesting case when different criteria are identified as the most sensitive according to the two cases. Additionally, it is clear that the order of preference for the top 5 alternatives is less sensitive to alterations in the original criteria weights compared to maintaining the order of preference of all 27 alternatives as the maximum and minimum allowable alterations to the criteria weights are significantly larger. The weight stability intervals will always be larger, or at least equal, when seeking to maintain the order of a subset of alternatives compared to maintaining the full order of all alternatives. This always occurs because any consideration of the top 5 alternatives (or indeed any of the top $q < n$ alternatives for that manner) is also considered when all 27 alternatives are examined.

5.2.4. Sensitivity case 3

Pareto dominance may be used to identify the most sensitive criterion/criteria weight when **both** the allowable absolute relative increase and decrease in a criterion weight are to be considered (Case 3). Based on Fig. 3A it may be deduced that criteria C4 (Feedstock total solids) and C5 (Energy density of AD plant) are the Pareto dominant criteria from a sensitivity perspective when both an increase and decrease in criteria weight are considered when aiming to maintain the order of preference of all 27 alternatives. On the other hand, criterion C5 (Energy density of AD plant) is the most sensitive criterion when aiming to maintain the order of preference of the top 5 alternatives when the allowable relative increase and decrease in criteria weights are considered (C5 is the Pareto dominant point in Fig. 3B).

In each event, it may be possible to use the WSM method itself to identify the most sensitive criterion/criteria weight. If the allowable relative increase and relative decrease in a criterion weight are of equal importance to the decision maker, applying the WSM method to the non-dominated criteria would yield the same result as Case 2 and criterion C5 (Energy Density of the AD Plant) would be the most sensitive criterion. If the allowable relative increase and decrease in a criterion weight are not of equal importance, then the most sensitive criterion according to the WSM method would be either C4 (Feedstock total solids) or C5 (Energy density of AD plant) depending on the level of importance assigned to the allowable increase or decrease in criterion weight if seeking to maintain the order of preference of all 27 alternatives.

When aiming to maintain the order of preference of the top 5 alternatives, applying the WSM method to the results presented in Fig. 3B will result in criterion C5 (Energy density of AD plant) being selected as the most sensitive criterion regardless of the relative importance of the allowable percentage increase or decrease in the criterion weight as it is the only non-dominated point.

As such, the sensitivity of the original WSM result to changes in the criteria weights when considering the allowable percentage increase and decrease in weight values can be facilitated through the reapplication of the WSM method to the non-dominated criterion/criteria. If there is only one non-dominated criterion it will always be the most

sensitive criterion regardless of the relative importance of the allowable percentage increase or decrease in criteria weight. If there are more than one non-dominated criteria, then the most sensitive criterion will depend on the relative importance of the allowable percentage increase and decrease to the criteria weights.

5.2.5. Managerial implications, applications to other MCDA methods, and further developments

The WSM method is one of the most widely used aggregation methods within MCDA in academic literature and in practice owing to the relative simplicity of the method in terms of implementation. The results of any MCDA should be subject to scrutiny by assessors and decision makers (such as managers within organisations). Within this process of scrutiny, the criteria weights are often a point of discussion as alterations to the weights can alter the ranking of alternatives. As outlined in Section 2 the most commonly used methods to assess the sensitivity of MCDA results to alterations in criteria weights include manual iterative alterations to each criterion weight, and the use of different sets of criteria weights in a form of scenario analysis. This process can be time consuming owing to the potentially large number of alterations to criteria weights required, and can also be inaccurate as the precise criteria weights required to cause a change in ranking are not determined. More refined methods which rely on stochastic modelling, or the use of MILP methods can provide greater accuracy, however, these can be computationally expensive, and the use of commercial software may present a barrier to some users and MCDA practitioners. The methodology developed in this paper to precisely calculate weight stability intervals and conduct three forms of sensitivity analysis avoids the need for the manual alteration of criteria weights (which can save time) and provides accurate results (thereby increasing confidence in the analysis). The ultimate benefit to decision makers is an increase in the robustness of the MCDA completed whilst saving time and increasing the accuracy and reliability of the sensitivity analysis.

The sensitivity analysis proposed in this work is based on a methodology to calculate the precise weight stability intervals for criteria weights used with the WSM aggregation approach. As outlined in the Introduction there is a plethora of MCDA methods in the literature including ELECTRE, compromise programming, SMART, AHP, TOPSIS, PROMETHEE, stochastic multi-objective acceptability analysis, and VIKOR. The use of weight stability intervals as a means of sensitivity analysis has been proposed for use with the PROMETHEE method developed by (Mareschal, 1988), however, the application of the WSI methodology developed in this work to other aggregation methods commonly used in MCDA such as compromise programming and TOPSIS remains unaddressed. As a first step, based on the fact that the compromise programming method and TOPSIS both rely on the use of L_p metrics to determine the “distance” to the utopia or nadir solution, the methodology developed in this work for calculating precise weight stability intervals for the WSM aggregation method should be applicable to compromise programming and TOPSIS with some modifications and should be pursued in further work. Pursuant to this, the application of modified WSI calculation methods should be investigated for other MCDA methods such as the Characteristic Objects Method (COMET) (Salabun, 2015) and Stable Preference Ordering Towards Ideal Solution method (SPOTIS) (Dezert et al., 2020). A further area for future research could be the modification and application of the developed method to calculate weight stability intervals for local weights, such as those proposed in the work by (Więckowski et al., 2022). Additionally, the implementation of the methodology developed in this work to calculate weight stability intervals in an explicit manner (that does not rely on enumeration or an iterative alteration of weights) and the use of these stability intervals in sensitivity analysis within a software package similar to PySensMCDA (Paradowski et al., 2024) would enable the dissemination and widespread use of said methods in an easy to use format.

6. Conclusion

Prior works have allowed for the calculation of weight stability intervals using the PROMETHEE method and identification of the most sensitive criterion using the WSM aggregation method within MCDA, respectively. Literature in which the weight stability intervals for the WSM (L_1) aggregation method is calculated using explicit analytical solutions that do not rely on enumeration has remained undeveloped until now. The use of scenario analysis to assess the impact of weight alteration is widespread in literature associated with the WSM aggregation method and exact analytical methods such as those developed in this work are absent. The methodology outlined herein allows for the precise and exact calculation of weight stability intervals for the original un-normalised weights and the normalised weights used in the WSM method to maintain the full ranking of alternatives, or the ranking of a subset of alternatives. This methodology does not rely on the use of complex MILP methods. The methodology also allows for the identification of the most sensitive criterion with regards to the minimum absolute percentage change in a criterion weight, the minimum absolute difference between the allowable percentage increase and percentage decrease in a given criterion weight, and through the application of a Pareto dominance approach when considering both the absolute relative increase and decrease in criteria weights. This can enable decision makers to focus their resources and time on refining the weights associated with the most sensitive criterion or a group of sensitive criteria.

It is hoped that the methodology developed and documented in this work will facilitate the calculation of weight stability intervals and their use in the identification of the most sensitive criteria weights used in the WSM method in a more nuanced and accurate manner. The managerial implications of this methodology include an increase in the robustness of the analysis whilst saving time and increasing the accuracy of the weight sensitivity analysis.

Future research on the topic of closed form solutions for sensitivity analysis may be extended to other aggregation steps in distance-based compensatory MCDA methods including compromise programming (which uses the L_2 , Euclidian distance) and The Order of Preference by Similarity to the Ideal Solution (TOPSIS). Future work on the use of the proposed method to calculate weight stability intervals for local weights should also be pursued. Incorporating stochastic modelling, whereby criteria weights are randomly varied within their respective stability intervals could be assessed in future work to add to the robustness of this approach. The future development of software packages and tools which enable to application of the method developed in this work would also enable more precise sensitivity analysis in future works. Finally, a deeper investigation into the concept of sensitivity and its definition may also be required which could make use of the WSM aggregation method to identify the most sensitive criterion weight from the non-dominated criteria weights.

CRediT authorship contribution statement

Richard O'Shea: Conceptualization, Formal analysis, Investigation, Methodology, Visualization, Writing – original draft, Writing – review & editing. **Peter Deeney:** Conceptualization, Formal analysis, Writing – review & editing, Project administration. **Evangelos Triantaphyllou:** Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Luis Diaz-Balteiro:** Formal analysis, Methodology, Writing – review & editing. **S. Armagan Tarim:** Formal analysis, Investigation, Writing – review & editing.

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Declaration of competing interest

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Appendix A. Supplementary data

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Appendix B. Supplementary data

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Data availability

Data will be made available on request.

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